

Chapter 1 Introduction to Fluid Mechanics

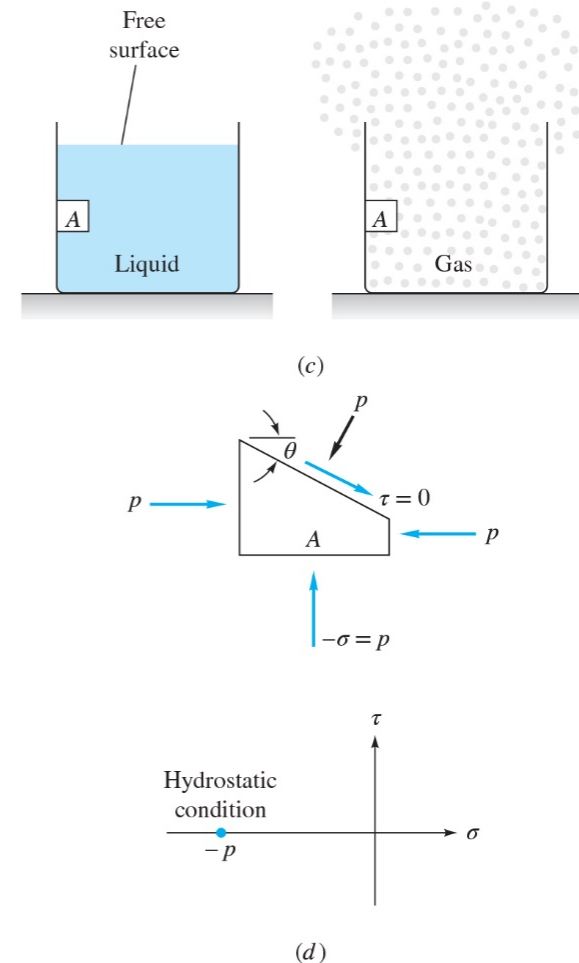
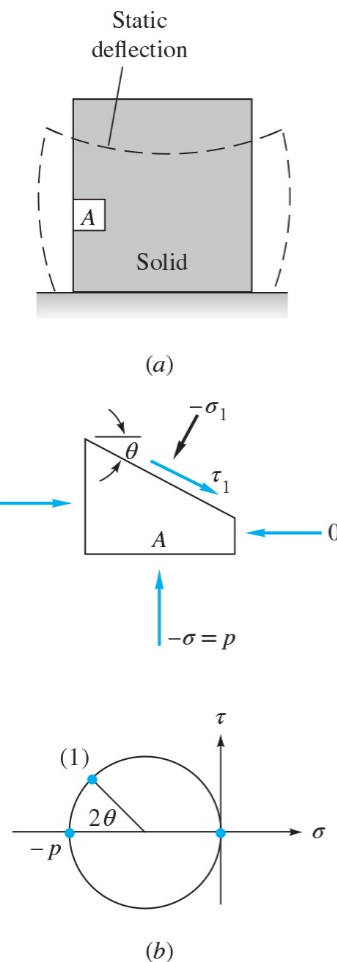


Learning Objectives

- Concept of fluid.
- Fluid as a continuum.
- Dimensions and units in fluid mechanics.
- Control volume and control surface for fluids flow.
- Eulerian as a method of description for most of fluid flow applications.
- Viscosity and other properties of fluid and flow.
- Three basic flow analysis techniques used in fluid mechanics.

The Concept of a Fluid

- Fluid Mechanics consists of two subjects: *Fluid Statics* and *Fluid Dynamics*.
- Both gas and liquids are classified as fluids.
- Solid can resist a shear by a static deformation.** (Fig 1.1 (a) and (b))
- Fluid cannot resist shear.** (Fig 1.1 (c) and (d))

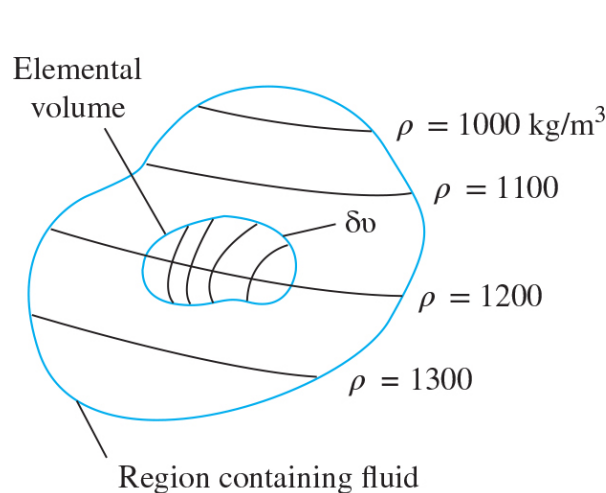


The Fluid as a Continuum

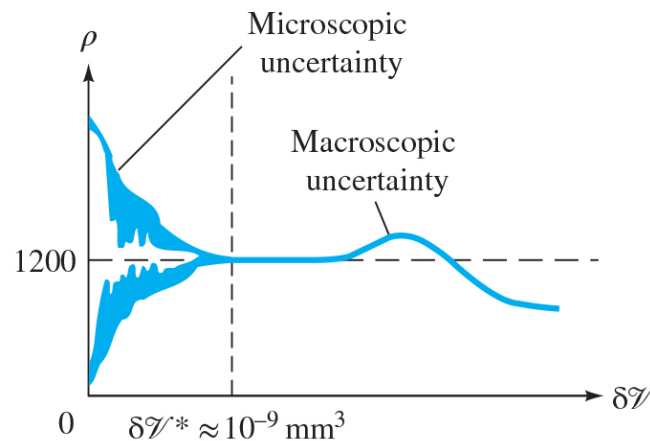
- **Fluids are aggregations of molecules:** widely spaced for a gas, closed spaced for a liquid. The distance between molecules are much larger than with the molecular diameter.
- **The fluid as a continuum.** The density of a fluid ρ is defined as:

$$\rho = \lim_{\delta v \rightarrow \delta v^*} \frac{\delta m}{\delta v}$$

The limiting volume v^* is about 10^{-9} mm^3 for all liquids and gases at atmospheric pressure.



(a)



(b)

Dimensions and Units

- There are only four *primary dimensions* from which all other dimensions can be derived: mass, length, time, and temperature.

Primary dimension	SI unit	BG (British Gravitational) unit	Conversion factor
Mass {M}	Kilogram (kg)	Slug	1 slug = 14.5939 kg
Length {L}	Meter (m)	Foot (ft)	1 ft = 0.3048 m
Time {T}	Second (s)	Second (s)	1 s = 1 s
Temperature { θ }	Kelvin (K)	Rankine ($^{\circ}\text{R}$)	1 K = 1.8 $^{\circ}\text{R}$

Dimensions and Units

- In the *SI system*, the *basic units* are kilograms $\{M\}$, meters $\{L\}$, seconds $\{T\}$, and temperature $\{\theta\}$. They are referred as MLT θ system of dimension. Force has secondary dimensions expressed $\{F\} = \{MLT^{-2}\}$. It can be derived from $\mathbf{F} = m\mathbf{a}$.
- In the *BG system*, the basic units are pound-force $\{F\}$, feet $\{L\}$, seconds $\{T\}$, and temperature $\{\theta\}$. They are referred as FLT θ system of dimension. Mass has secondary dimensions expressed as $\{M\} = \{FT^2L^{-1}\}$.

The Principle of Dimensional Homogeneity

- All equations must be *dimensionally homogeneous*---each additive term in an equation must have the same dimension.

Example: Bernoulli equation for a steady, incompressible flow can be expressed as $p + \rho V^2 / 2 + \rho g z = \text{constant}$. Show that the equation satisfies the principle of dimensional homogeneity.

Solution:

We can examine the dimension homogeneity. Each and every term in this equation has dimensions of pressure $\{ML^{-1}T^{-2}\}$.

- What is the dimension for the constant in the equation?

- Theoretical equations in fluid mechanics such as the Bernoulli equation are dimensionally consistent and require no further conversion factors when these two fundamental unit systems are used.

System and Control Volume

- A **system** is defined as an arbitrary quantity of mass of fixed identity. It is also called closed system or *control mass*. Everything external to the system is denoted by the term surroundings. The system is separated from the surroundings by its boundaries.

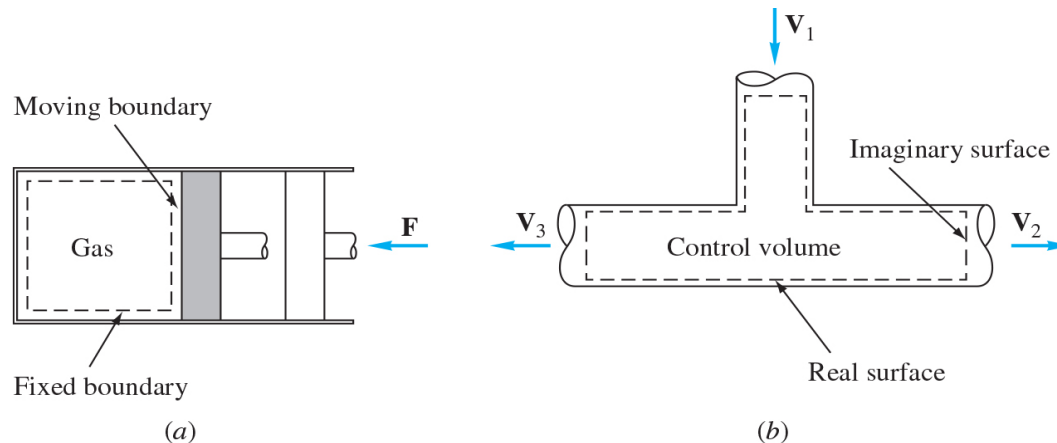


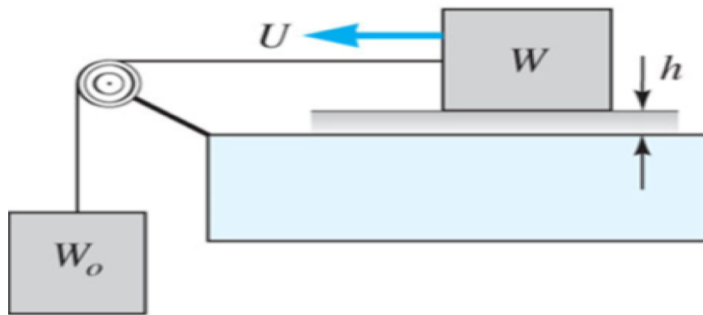
Fig.1.3 System and control volume.

- A **control volume** is defined as an arbitrary region in space through which fluid flows. **The control volume can be at rest or in motion** (Fig.1.3a). The geometric boundaries of the control volume separating it from the surrounding is called **control surface**. **The control surface can be real or imaginary** (Fig. 1.3b).

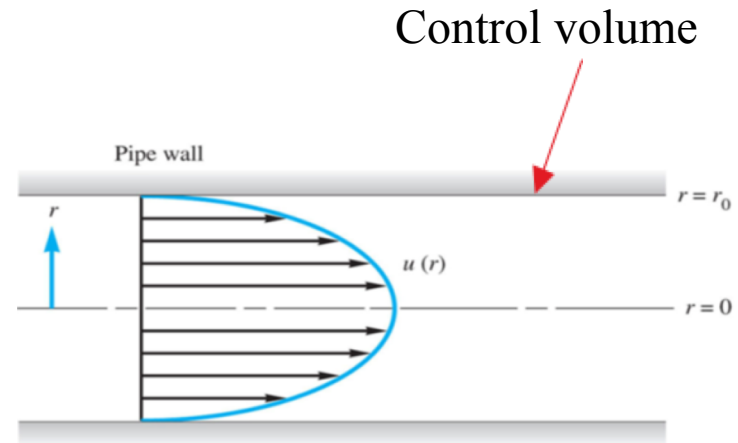
Method of Description

- **Lagrangian**: a frame of reference that follows the trajectories of individual particles or bodies.

Examples:



Analyze the position, velocity, and acceleration of the Block W



Analyze velocity, acceleration, and pressure fields for the fluid flow in a pipe

- **Eulerian**: the properties of a flow field are described as functions of space coordinates and time. As examples,

Velocity field: $\mathbf{V}(x, y, z, t) = \mathbf{i}u(x, y, z, t) + \mathbf{j}v(x, y, z, t) + \mathbf{k}w(x, y, z, t)$

Acceleration field: $\mathbf{a}(x, y, z, t) = \frac{d\mathbf{V}}{dt} = \frac{\partial \mathbf{V}}{\partial t} + u \frac{\partial \mathbf{V}}{\partial x} + v \frac{\partial \mathbf{V}}{\partial y} + w \frac{\partial \mathbf{V}}{\partial z}$

Thermodynamics Properties of a Fluid

- Pressure: compression stress at a point of static fluid. **Pressure difference is often the driving force for internal flow like pipe flow.**
- Temperature: indication of the internal energy level of a fluid. Temperature difference results in *heat transfer*.
 $^{\circ}\text{R} = ^{\circ}\text{F} + 459.69, \text{K} = ^{\circ}\text{C} + 273.16$
- Specific weight: fluid weight per unit volume. $\gamma = \rho g$ (N/m^3 or lbf/ft^3)
- Specific gravity: ratio of a fluid density to a standard reference fluid, usually water at 4°C for liquids and air for gases.

$$\text{SG}_{\text{gas}} = \frac{\rho_{\text{gas}}}{\rho_{\text{air}}} = \frac{\rho_{\text{gas}}}{1.205 \text{ kg/m}^3} \quad \text{SG}_{\text{liquid}} = \frac{\rho_{\text{liquid}}}{\rho_{\text{water}}} = \frac{\rho_{\text{liquid}}}{1000 \text{ kg/m}^3}$$

Giving the specific gravity of a fluid is equivalent as giving the density.

- Total energy: sum of internal energy, kinetic energy, and potential energy. In fluid mechanics, the total energy per unit mass is:

$$e = u + \frac{1}{2} V^2 + gz \quad (\text{kJ/kg})$$

State Relations

- **Gas**: the simplest state relation for gases at relative high temperature and low pressure (compared with the critical condition) is the *ideal gas law*

$$p = \rho RT \quad \text{where gas constant } R = c_p - c_v$$

$$k = \frac{c_p}{c_v} \quad \text{specific heat ratio}$$

For air as an ideal gas, $k \approx 1.4$ so

$$c_v = \frac{R}{k-1} = 4293 \text{ ft}^2/\text{s}^2 \cdot \text{R} = 718 \text{ m}^2/\text{s}^2 \cdot \text{K}$$

$$c_p = \frac{kR}{k-1} = 6010 \text{ ft}^2/\text{s}^2 \cdot \text{R} = 1005 \text{ m}^2/\text{s}^2 \cdot \text{K}$$

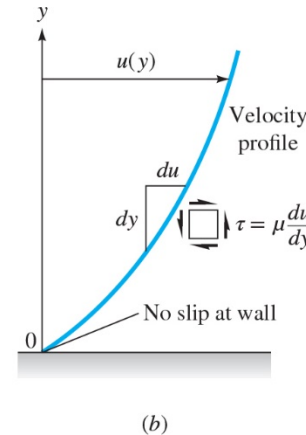
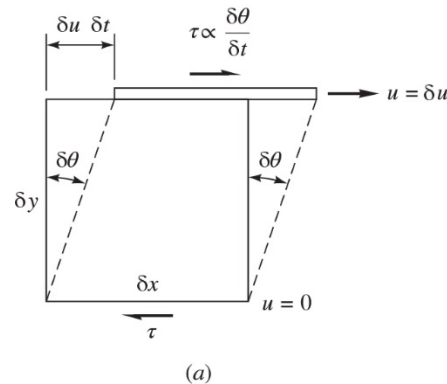
- **Liquid**: nearly incompressible. This implies that:

$$\rho = \text{const} \quad c_p = c_v = c \quad du = dh = cdT$$

In fluid dynamics, gaseous flow at low speed, where the velocity is much less than its speed of sound, can also be treated as incompressible flow.

Viscosity and Other Properties

- **Viscosity**: a quantitative measure of a fluid's resistance to flow. It determines the fluid strain rate under a given shear stress between the fluids.



For continuous shear deformation in a fluid,

$$\tau \propto \frac{\delta\theta}{\delta t} \rightarrow \tau = \mu \frac{du}{dy} \quad \text{since } \tan \delta\theta = \delta\theta = \frac{\delta u \delta t}{\delta y}$$

Fluids that follow the linear relation are called **Newtonian fluids**.

- **The Reynolds number**: ratio of inertial force over viscous force.

$$\text{Re} = \frac{\rho V L}{\mu} = \frac{V L}{\nu} \quad \text{where } \nu = \frac{\mu}{\rho}$$

Viscosity and Other Properties

Example: A block of weight W slides down on an inclined plane while lubricated by a thin film of oil. Assuming linear velocity distribution, find the terminal velocity V .

Solutions:

Let x coordinate be down the incline.

By “terminal” velocity, we mean no acceleration.

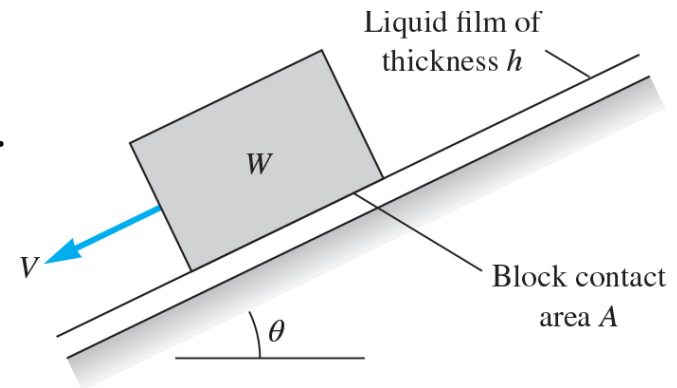
The force balance in x direction,

$$\sum F_x = W \sin \theta - \tau A = ma_x = 0$$

Assuming linear velocity distribution,

$$W \sin \theta - \left(\mu \frac{V}{h} \right) A = 0$$

$$\rightarrow V_{\text{terminal}} = \frac{hW \sin \theta}{\mu A}$$



Variation of Viscosity with Temperature

- **Gas**: viscosity increases with temperature. Two common approximations are:

$$\frac{\mu}{\mu_0} \approx \begin{cases} \left(\frac{T}{T_0}\right)^n & \text{Power law} \\ \frac{(T/T_0)^{3/2} (T_0 + S)}{T + S} & \text{Sutherland law} \end{cases}$$

where μ_0 is a viscosity at absolute temperature T_0 (usually 273K).

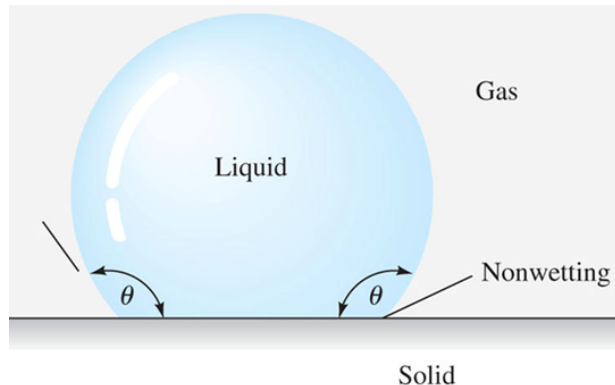
- **Liquid**: viscosity decreases with temperature. Common estimation is

$$\ln \frac{\mu}{\mu_0} \approx a + b \left(\frac{T}{T_0}\right) + c \left(\frac{T}{T_0}\right)^2$$

For water, with $T_0 = 273.16 \text{ K}$, $\mu_0 = 0.001792 \text{ kg / (m.s)}$, suggested values are $a = -1.94$, $b = -4.80$, and $c = 6.74$.

Surface Tension

- Phenomenon: A liquid, being unable to expand freely, will form an *interface* with a second liquid or gas.



When the liquid-vapor interface meets a solid surface, the *contact angle* θ measures the wettability of the solid surface by liquid. If $\theta < 90^\circ$, the liquid “wets” the solid, the surface is hydrophilic; If $\theta > 90^\circ$, the liquid is non-wetting, the surface is hydrophobic.

Surface Tension

- *Surface tension*: shown in Fig. 1.12, equal and opposite forces of are exposed normal to the cut.

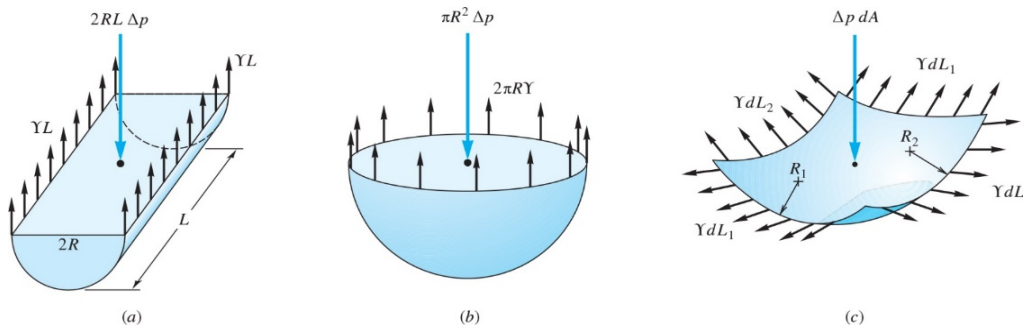
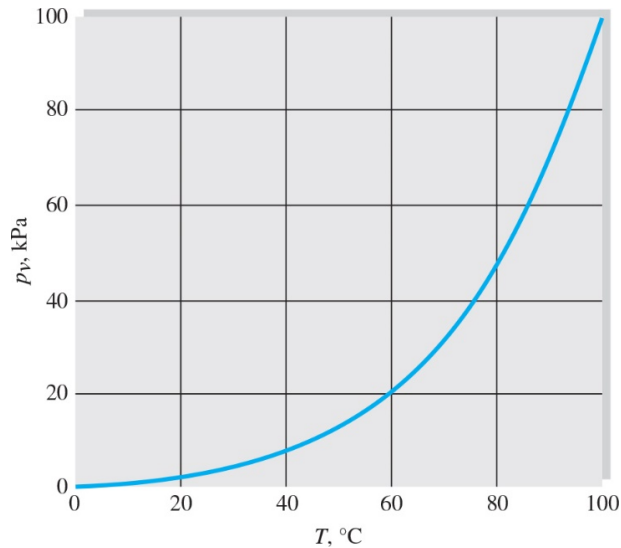


Fig.1.12 Pressure change across a curved interface due to surface tension.

Vapor Pressure

- **Vapor pressure**: the pressure at which a liquid boils and is in equilibrium with its own vapor.



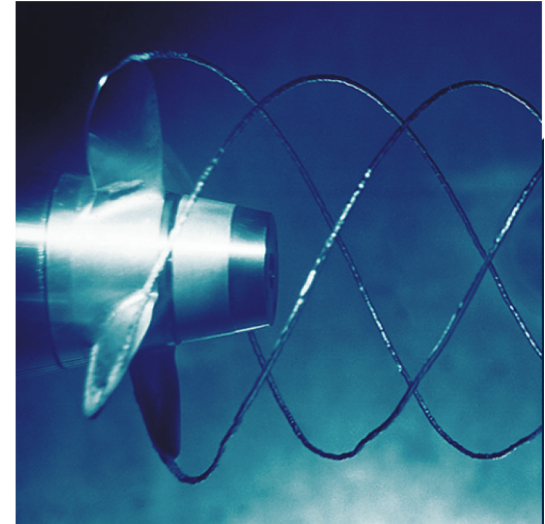
Example:

From the figure, the vapor pressure of water at 20°C is 2.34 kPa or 17.5 mmHg.

Fig.1.14 Vapor pressure of water.

Vapor Pressure

- **Cavitation**: The liquid vapor is dropped below the vapor pressure due to a flow phenomenon.



Courtesy: Penn State, APL

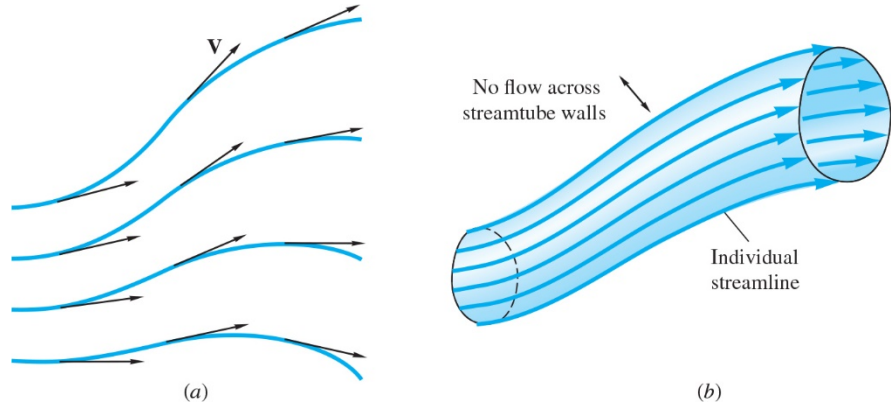
Example:

If water is accelerated in a pump from rest to about 50 ft/s, its pressure drops by about 15 lbf/in², or 1 atm (~103 kPa). This can cause cavitation.

Flow Patterns

- **Streamline**: a line everywhere tangent to the velocity vector at a given instant.

Fig.1.17 (a) streamlines are everywhere tangent to the local velocity vector; (b) a streamtube is formed by a closed collection of streamlines.



Since $d\mathbf{r}$ (dx, dy, dz) of a streamline is parallel to $\mathbf{V}$ (u, v, w),

$$\text{Streamline equation: } \frac{dx}{u} = \frac{dy}{v} = \frac{dz}{w} = \frac{dr}{V}$$

- **Pathline**: actual path traversed by a given fluid particle.
- **Streakline**: locus of particles that have passed through a prescribed point.

Streamlines, pathlines, and streaklines are identical in steady flow.

Basic Flow Analysis Techniques

- For solid mechanics: use the *free-body diagram* extensively for the analysis. But for fluid mechanics: use a system or a control volume, depending on the problem being analyzed.
- In all cases, the flow must satisfy the three basic laws of mechanics plus a state relation and associated boundary conditions. The expressions of these basic laws in a control volume are different.
 - Conservation of mass, (linear or angular) momentum, and energy
 - Equation of state
 - Associate boundary conditions

Basic Flow Analysis Techniques

- **Integral approach**: formulating the basic laws of fluid mechanics in a finite region, making a balance of flow in versus flow out, and determining gross flow effects such as the force or torque on a body or the total energy exchange. (Chapter 3)
- **Differential approach**: formulating the basic laws of fluid mechanics in an infinitesimal control volume, and seeking to describe the detailed flow pattern at every point (x, y, z) in the flow fluid. (Chapter 4)
- **Dimensional analysis** and *experimental approach*: conducting dimensional analysis based on the Pi theorem, and providing insight for experimental study. (Chapter 5)

Computational Fluid Dynamics (CFD) has become a common engineering tool for a large class of fluid flow, heat, and mass transfer problems.

The History of Fluid Mechanics

- A history of erratically occurring early achievements.
- Era of steady fundamental discoveries in the 18th and 19th centuries.

Name	Important Contribution
Archimedes (285–212 BC)	Established laws of buoyancy and floating bodies.
Leonardo da Vinci (1452–1519)	Formulated the first equation of continuity.
Isaac Newton (1642–1727)	Postulated the law of linear viscous stresses.
Leonhard Euler (1707–1783)	Developed Bernoulli's equation by solving the basic equations.
L. M. H. Navier (1785–1836)	Formulated the basic differential equations of viscous flow.
Jean Louis Poiseuille (1799–1869)	Performed first experiments on laminar flow in tubes.
Osborne Reynolds (1842–1912)	Explained the phenomenon of transition to turbulence.
Ludwig Prandtl (1875–1953)	Formulated boundary layer theory, predicting flow separation.
Theodore von Kármán (1881–1963)	Major advances in aerodynamics and turbulence theory.

Summary

- A fluid, unlike a solid, must move if subjected to a shear stress.
- Under Eulerian description, the properties of a flow field are described as functions of (x, y, z, t) .
- You must learn to locate and use all following properties to become proficient in fluid mechanics. They include the pressure p , density ρ , and temperature T , and many secondary properties such as viscosity μ , specific weight γ , surface tension Y , and vapor pressure p_v .
- There was a brief discussion of the five different kinds of mathematical relations we will use to solve flow problems---mass conservation, linear momentum, energy conservation, equations of state, and appropriate boundary conditions at walls and other boundaries.
- Flow patterns are also discussed briefly. The most popular, and useful, scheme is to plot the field of streamlines, that is, lines everywhere parallel to the local velocity vector.